

Transportation 5.0: The DAO to Safe, Secure, and Sustainable Intelligent Transportation Systems

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Abstract—In 2014, IEEE Intelligent Transportation Systems Society established a Technical Committee on Transportation 5.0 with the mission of promoting and transforming the deployment of advanced and innovative technologies, especially Artificial Intelligence in transportation. This paper briefly summarizes our main research and findings over the last decade. Transportation Foundation Models, Transportation Scenarios Engineering, and Transportation Operating Systems have been identified as the main directions for the research and development of next-generation intelligent transportation systems.

Index Terms—Intelligent transportation systems, cyber-physical-social systems, parallel intelligence, knowledge automation, DAOs, transportation 5.0.

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I. TRANSPORTATION: A SOCIO-TECHNICAL SYSTEM

THE relationship between Intelligent Transportation Systems (ITS) and human societies is multifaceted, encompassing social morphology, transportation system design, and social behavior, all of which are interdependent and influence one another. Society plays an equal or even more significant role than transportation technologies in the development and evolution of ITS [1], [2]. In established facilities, such as intelligent surveillance and traffic light control, ITS has utilized societal and transportation-related human factors as contextual inputs, leading to a reactive response. The neglect of societal perspectives in previous studies was not due to a lack of recognition of their importance, but rather because of the tremendous complexity involved in accounting for social interactions. However, with the emergence of the Intelligent Technology Era (the New IT Era), the situation has changed.

A recent methodology known as Transportation 5.0 has been proposed with a Cyber Physical Social Systems (CPSS) architecture [3]. Inspired by Karl Popper’s theory of reality [4], CPSS refers to the integration and coordination of three elements: computational (or cyber) resources, physical resources, and human or social traits [3], [5]. It emphasizes the deep integration of social and human dimensions into Cyber Physical Systems (CPS), establishing a new space with the same status as cyber and physical spaces, rather than being confined as a mere backdrop or context.

Decentralized Autonomous Organizations and Operations (DAO₂) have opened up a new world with new social norms and production relationships. Interestingly, the well-known ancient Chinese philosopher Lao Tzu (6th-4th century BC) started his classic Tao Teh Ching with six Chinese characters: “道(Dao) 可(Ke) 道(Dao) 非(Fei) 常(Chang) 道(Dao)”, Dao means journey or road in Chinese, inspires the incorporation of social factors in transportation systems [6]. By incorporating DAO into CPS, Transportation 5.0 is capable of treating society as an active entity, proactively interacting with society in ITS processes by leveraging advancements in Artificial Intelligence (AI), large-scale communication and computing, and data technologies.

From the perspective of social operation and urban development, the transportation system encompasses all the systems that facilitate the movement of people and goods within a city.

The level of urban development is highly reflected in the quality of a city's transportation system. Likewise, the transportation system significantly impacts the regional economic structure and urban morphology [7]. The limited capacity and functionality of transportation systems can constrain the growth potential of cities.

Although ITS technologies have experienced significant attention and growth, optimizing a transportation system necessitates more than just technological advancements. In fact, new technologies may introduce additional complexities to existing systems, which may lead to inefficiencies and degraded performance if not properly designed and optimized. The current transportation system is a collection of many uncoordinated subsystems each of which is trying to optimize its performance without caring about the overall system performance leading to well-known phenomenon of Price of Anarchy [8] where subsystem optimality does not imply system optimality and vice versa. The development of personal transportation, for example, runs counter to the need for social welfare due to the high consumption of roadway resources that results from the increasing demand for personal transportation [9], [10]. As a result, past transportation planning objectives, which focused on ensuring efficient private transportation in decades, have led to severe environmental pollution and traffic congestion [11].

Several everyday occurrences emphasize the significance of taking a societal perspective into account when designing transportation systems. For instance, social activities have become a primary purpose of trips for many people. Despite the COVID pandemic, the proportion of trips involving visiting friends and dining out went up from 48.6% in 2018 to 52.6% in 2020 in Beijing [12], [13]. In line with this, in light of the global COVID-19 pandemic, the importance of having a dependable and trackable logistics network became more critical than ever before [14].

Studies demonstrate the crucial role that humans played in all technological fields [15], particularly in fields where ITS has a significant impact, such as vehicle technologies and level of automation, traveler information management, fleet planning, and traffic management control. Societal perspectives should be prioritized early in the ITS testing phase to introduce a new ITS paradigm. The results of testing could lead to wider acceptance of the new ITS paradigm by the public. The increased acceptance of ITS could result in the improvement of its morphology, leading to a cycle of enhanced public awareness and continuous support of ITS developments. For example, the adoption of new technologies such as autonomous vehicles will depend on the public's trust in the technology [16]. Even if the introduction of these vehicles into urban environments is expected to be gradual, the actions of the road users and their reactions to the vehicles need to be investigated to understand the mechanisms to establish such trust [17].

During the 17th IEEE International Intelligent Transportation Systems Conference (ITSC) held in Qingdao, China, in 2014, the Technical Committee on Transportation 5.0 was established by the IEEE Intelligent Transportation Systems Society (ITSS). Since then, IEEE ITSC has regularly organized

workshops on Transportation 5.0 and published a number of technical reports [18], [19] and papers on topics of X 5.0 and Parallel Intelligence [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32]. This position paper serves as a brief summary of these extensive activities and research over the past decade.

The remainder of this paper is organized as follows: in Section II, we review the history of transportation systems through the lens of socio-technical systems. In Sections III and IV, we highlight the challenges brought about by recent social changes and the cornerstone technologies that could solve them. Section V then demonstrates best practices and upcoming ITS applications. Section VI concludes the paper.

II. A BRIEF HISTORY OF TRANSPORTATION SYSTEMS

A. Socio-Technical Transition

Socio-technical systems (STS) refer to systematic arrangements that comprise cybernetic processes, technical objects, and human relationships. These intricate infrastructures are composed of social and technological factors that interact in a cause-and-effect mechanism and in an unpredictable manner as well. STS incorporates the four critical elements: humans, nature, artifacts, and organizations. Humans interact with nature, which results in the creation of artifacts, which in turn contribute to the formation of organizations [33], [34].

Transportation can be classified as a typical example of STS. In this system, humans mainly interact with physical space and raw materials, which gives rise to human-made objects like roads and vehicles. During this process, humans not only construct the transportation instruments but also organize the operations involved, such as logistics and public transport services, which constitute an organizational action [35].

Previous research showed that the unilateral optimization of society or technology tends to increase the number of unpredictable, "non-designed" relationships and those relationships that are detrimental to the system's performance [36], [37]. Thus, the socio-technical theory is about joint optimization or designing social and technological systems to work together smoothly [38].

However, joint optimization is often difficult to achieve in socio-technical systems, because the optimization process depends on synchronized changes in both technological development and social awareness. When technological development exceeds the rate of change in social awareness, such development is often suppressed. Similarly, when social awareness exceeds the level of technological development, such social awareness is often limited to a small scope. Therefore, for complex systems such as transportation, it is necessary to use a socio-technical systems perspective to study them. This is the only way we can understand how to choose the appropriate path to adapt to the development of social awareness.

B. Evolution of Transportation Systems

By using the lens of socio-technical paradigm shifts, we can gain a deeper understanding of travel as a fundamental human

need and divide the evolution of the transportation system into four phases. These four phases reflect the general pattern of socio-technical systems development, whereby advances in tools and technologies first bring about changes in societal forms, which in turn, begin to guide technological change in a reverse direction.

Transportation 1.0: Transformation of the natural environment. The emergence of cities and nations led to the emergence of transportation systems. When the scope of human and commodity activities expanded beyond traditional tribal societies, the development of new transportation methods and the construction of transportation infrastructure became a fundamental requirement for the development of cities and nations [39]. The country's road networks, such as the road network of Rome, the Qin Zhi Dao of China, and the Persian Royal Road, have been constructed, allowing the natural environment to support primitive transportation. More importantly, travel as a social phenomenon became familiar to humans, and the regions connected by the transportation system became essential to national identity.

Transportation 2.0: Proliferation of transportation means. As familiarity with travel grew, the evolution of travel methods became a significant aspect of the 2.0 era. This development was initially driven by the ruling class, which resulted in the creation of primitive modes of transportation such as saddles, carriages, and sailing ships [40]. With the advent of the industrial revolution, modern tools such as automobiles, trains, and ships were created. The introduction of the Ford assembly line marked the peak of the second industrial revolution, as well as the Transportation 2.0 era. Since then, millions of people have had access to their own cars, and travel, a privilege in the 1.0 era. With the advent of the automobile age, it has become impossible to conceive of a society without vehicles for travel. The transportation system has become both a spatial foundation and an essential component of the modern city [41], [42].

Transportation 3.0: Organizing the mobilities. With the ubiquity of transportation, the social coordination of mobilities has become an indispensable component, enabling the smooth and efficient operation of marine, air, rail, and urban road transportation, respectively [43], [44]. In the 3.0 era, travel resource coordination has become the research focus. The primary outcomes of these studies are the standardization of transportation means and developing relevant laws and regulations. One after another, standard-gauge railway, International Regulations for Preventing Collisions at Sea (COLREGs) [45], the Convention on International Civil Aviation [46], and other conventions have been proposed. The most well-known of these conventions is the Vienna Convention on Road Traffic and Road Signs and Signals [47], which regulates nearly all aspects of modern urban transportation.

Transportation 4.0: Fulfilling the human need. As transportation systems have become increasingly important in our daily life, the focus of transportation system research has shifted back to its original goal: to fulfill human needs. This shift in research focus is fundamentally similar to all socio-technical systems, stemming from a misallocation of resources

compared to the resource demand [48], [49]. Transportation is fundamentally a social need, and the vast majority of travel demand stems from social needs such as commuting and other social activities. However, the primary owners of transportation resources are individuals or businesses whose primary goal in providing resources is to satisfy individual interest claims rather than social needs, squeezing the resource for public transportation. Furthermore, transportation system development has noticeable and often negative effects on other social systems, such as personal safety and environmental protection [50]. Transportation 4.0 research focuses on better aligning transportation systems with human needs. Transportation safety and protection measures, such as airbags and seat belts, make travel much less dangerous to personal safety. Using new energy vehicles reduces the environmental impact of the transportation system. New modes of transportation, such as ride-sharing and on-demand service, reduce the need for private ownership of transportation resources, providing public transportation with the opportunity to improve its effectiveness in meeting its resource needs [51], [52].

C. Artificial Intelligence: The New Frontier

In conclusion, the development of transportation systems has experienced four paradigm shifts in its research focus on transforming nature, creating artifacts, enhancing organization and satisfying human desires. As a socio-technical system, the complexity of transportation systems has significantly increased from 1.0 to 4.0, making it challenging to achieve sufficient gains through technical optimization alone [53].

While tackling traffic issues, taking social structural changes into account can have surprisingly beneficial outcomes. For example, new business models such as bike-sharing have significantly impacted the efficiency of the traffic system, though it contains very few technical inventions. However, without a well-thought-out plan, such efforts may lead to unintended consequences. For instance, the rise in popularity of courier delivery services has increased the use of non-motorized vehicles, leading to a corresponding increase in the rate of accidents associated with them [54], [55]. As a result, if a better transportation system is desired, new approaches to combining nature, artifacts, organization, and people are required.

Fortunately, the advancement of AI technologies has made such convergence possible. The proliferation of intelligent technology has enabled the transformation of many previously human-responsive behaviors into the automatic behaviors of machines, making direct modification of social consciousness possible [56], [57]. As previously stated, the focus of the transportation system has shifted towards achieving efficiency in transforming nature, creating useful artifacts, constructing reliable organizations, and meeting human needs with trustworthiness. These four most essential aspects can be combined into one in the new Intelligent Technology era, allowing for the synergistic optimization of multiple needs and acceleration of the paradigm shift from technology-centered to society-centered, as illustrated in Fig. 1.

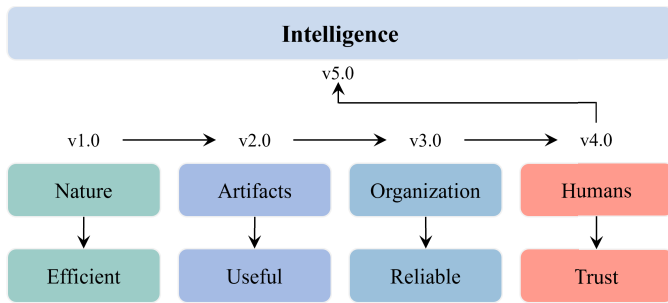


Fig. 1. The evolution of transportation systems through four distinct phases that align with paradigm shifts in socio-technical systems. Each phase reflects the interplay between societal changes and technological advancements.

III. DECENTRALISED SOCIETY (DESOC): TRENDS AND CHALLENGES

A. From New Technology to New Social Form

In the previous sections, we discussed how intelligent technology will facilitate the integration of natural entities, artificial artifacts, social organizations, and humans. In this section, we will discuss the global impact of technology and the potential forms of future societies.

The CPSS theory considers three worlds: the Physical World (World One), the Mental World (World Two), and the Artificial World (World Three), as an organic whole [4]. This perspective proposes a profound understanding of the laws of socio-technical system development. As illustrated in Fig. 2, changes initially occur in the artificial world, which subsequently affects the mental and physical worlds. The three worlds connected through the artificial world, namely, a parallel world, will eventually serve as critical foundations for future advancements. The artificial world we are facing today has undergone significant transformation leading to a consequential impact on our mental and physical surroundings. Recognizing these changes appropriately is a prerequisite for the advancement of transportation systems.

B. The Decentralization Trends

As new social forms are gradually being constructed, the development of intelligent technology serves as the catalyst, while the development of energy technology and communication technology respectively provides the material and informational exchange infrastructures. We believe that these three technological innovations have profoundly changed the human-made world and further shaped the present physical and mental world. Ultimately, society will evolve and develop towards a decentralized direction and trend.

- **Energy Technology.** The development of clean energy is considered as crucial technological advancement that relieves human-environmental tensions from fossil fuel consumption. Furthermore, it has accelerated the maturation of battery technology. Presently, the development and widespread usage of electric vehicles and smartphones are closely linked to innovations in battery technology [58], [59], [60]. The shift from the centralized supply of fossil fuels, such as coal and natural gas, to the decentralized supply of battery-based energy, reflects a

move towards decentralization in energy utilization. More importantly, this progress in energy technology provides a key enabling foundation for the development of digital technologies.

- **Communication Technology.** Communication technology is an indispensable aspect of modern civilization in the 21st century, with smartphones leading to extensive advancements [61], [62]. These changes have significantly impacted modern life. The widespread use of modern communication technologies like WiFi and 5G have integrated search engines and e-commerce into everyday life. Social networks and short videos have replaced traditional news sources and leisure activities [63], [64].
- **Intelligent Technology.** Web and communication services have driven the development of technologies such as recommendation engines, large-scale databases, and application-specific integrated circuits. These innovations have enabled the processing of large volumes of data and facilitated the creation of powerful computing infrastructures, which have fueled the development of AI technologies capable of performing various intelligent tasks [65], [66]. Deep learning, along with various new technologies, has dramatically increased automation in speech and image recognition as well as intelligent decision-making [67], [68], [69]. However, these advancements sometimes pose a challenge for humans to comprehend.

Initially, these technological advancements were developed to enhance human connectivity and increase efficiency. However, their development has often led to unforeseen consequences. While social networks and short videos have accelerated the spread of news, they have simultaneously exacerbated the problem of information cocoons and the fragmentation of opinions [70]. The widespread adoption of smartphones and online entertainment has made attention one of the most highly sought-after and in-demand commodities of our era. These technological advancements affect the physical world in addition to the mental world. Cities have become crowded with intelligent technology, including face-recognition cameras and even autonomous vehicles (AVs). Due to internet accessibility and remote work, the geographical distribution of the workforce underwent notable transformations [71].

C. The Challenges of New Social Form

The rise of the aforementioned technological innovations has led to societal fragmentation, with increasingly decentralized information channels, diverse controlling subjects, and pluralistic value orientations [72]. Bitcoin and other decentralized digital currencies, as well as Non-Fungible Tokens (NFTs), have gained immense popularity [73]. The most powerful subjects that influence society have shifted from city mayors to internet users who can post relevant news at the right time. Celebrities are no longer able to maintain their careers for decades, and instead, the concept of “15 minutes of fame” introduced by Andy Warhol has taken their place [74]. The phenomenon described above is often referred to as DeSoc, which stands for Decentralized

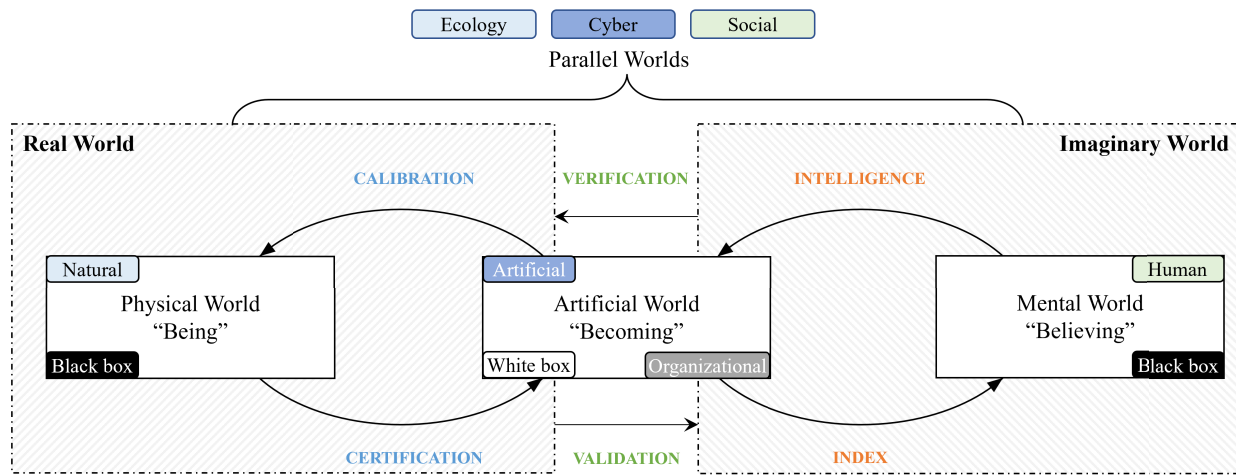


Fig. 2. Three worlds in socio-technical systems shows the interconnected nature of the Physical World (World One), Mental World (World Two), and Artificial World (World Three).

Society [75], [76]. Three of these changes in DeSoc, which have had a significant impact on the transportation system, will be examined in detail below.

1) *Decentralization of Information Channels*: In the era of Web 3.0, the development of mobile apps and IoT technology has enabled almost anything to be transformed into a viable information channel [77]. Theoretically, more comprehensive information sources could enhance traffic management systems' accuracy and efficiency [78]; however, data barriers such as data segregation due to privacy protection laws and competing relationships among systems impose significant limitations on this approach. These barriers obstruct information channels and substantially restrict the transportation system's ability to adjust dynamically to evolving conditions. Despite a considerable amount of data used to analyze and monitor traffic dynamics, data-driven decision-making technologies often have limited access to local data, and their potential to benefit society remains untapped.

The primary cause of data barriers is the lack of well-defined and properly implemented data pricing methods [79], [80]. A price that is deemed reasonable and equitable must be agreed upon by both data providers and consumers to facilitate collaboration. Current data pricing strategies are typically tailored for single-system information channels where a single data source is targeted for a particular scenario. This hinders the growth of a mature data market and ultimately leads to the existence of data barriers. Balancing the privacy needs of individuals and the concerns of businesses about the diminishing value of data as a result of privacy violations remains a challenging and unresolved issue. New industry chains, including data collaboration platforms and data as a service (DaaS) [81], should be indispensable tools for overcoming this barrier.

2) *Absence of Consensus*: Individuals often perceive and react to phenomena differently. In the past, the uniqueness of advertising channels resulted in society overlooking individual diversity and, therefore, lessening its influence on social interaction. The fragmentation of information channels has made the effects of diversity more apparent over time [82].

Nowadays, it is common for two contrasting mainstream views on significant social issues to surface, indicating that the current mechanism for forming a consensus is not suitable for current situations.

Building simulation systems is the typical solution to reaching consensus in social engineering. Participants can reach a consensus on the mechanism and response strategy of a social phenomenon by mathematically modeling it and fitting observations to it. In the domain of ITS, examples of consensus are traffic flow models, car-following models, lane-changing models, and human driver models created through simulation. Current traffic system simulations depend on rules made by experts that constitute a singular consensus. Designing a system in this way accommodates simple social systems with one data source and allows researchers and decision-makers to create a dataset to assess the effectiveness of traffic planning and control strategies. However, with the increasing abundance of traffic data, the disparities between simulated data and actual data are becoming more apparent. Furthermore, the increase in control devices has made algorithms relying on simulation systems less dependable.

To solve these problems, the existing consensus formation mechanisms need to be combined with varying perspectives based on the observer, encompassing a wider range of facets. The simulation mechanism should be automatically separated into several sub-simulations based on different observation perspectives and then altered using a data-driven, learning-based method. This shift is significant because it replaces a strongly consistent consensus with a mechanism for reaching local consensus on matters of common interest. For example, when utilizing a data-driven approach for a car-following model in a specific region, the simulation output will adjust to the area's data granularity and patterns. Such a paradigm will allow each district to present data in diverse formats, which is non-trivially different from the traditional method.

3) *Conflicting Interests*: In the transportation industry, conflicts have been resolved by researchers using macro-level statistics. For example, the crowding-out effect of private cars on public transportation has been identified, leading to

the creation of bus lanes [83]. However, in situations where information sources are widely distributed and consensus can only be reached at the local level, participants may develop conflicting interests. Drivers tend to criticize pedestrians who cross the street, but they also tend to complain about drivers who fail to yield when they are not driving. These disagreements are common, and in situations where there is competition for resources, divergent opinions on who should have the right to own or use limited resources or benefits are likely to arise.

While macro-level statistics have been used to resolve conflicts in the transportation industry, micro-level conflicts have also been identified through data collection technologies and control devices. For example, decision-makers in the transportation industry often choose to optimize arterial traffic volumes at the expense of non-arterial traffic when traffic data is limited, which can be effective. However, with the emergence of mobile apps that enable individuals to predict their arrival time, there is a risk of a significant surge of vehicles on the arterial path, which can undermine the signal optimization strategy [84].

The above examples demonstrate that when there is ample information available but no significant consensus on how to allocate limited resources or benefits, relying solely on macro-level statistics and control methods may not be sufficient to address complex social conflicts. Instead, a multidimensional control strategy must be developed to address the dynamic game characteristics of social behavior, especially when micro-level factors can significantly alter the expected outcome.

IV. TRANSPORTATION 5.0: THE DAO TO SAFE, SECURE AND SUSTAINABLE ITS

The advancement of technology, particularly the Internet and information technology, has led to decentralization in numerous aspects of society, such as information channels, societal consensus, and interest demands. This shift has significantly impacted the design, planning, and management of transportation systems, which have traditionally relied on experience, theory, and values.

Current centralized transportation systems often face numerous challenges in adapting to a decentralized society. Centralized transportation systems, which are normally designed and managed by a unified authority, assume that the public transportation demands are relatively uniform. On the contrary, in a decentralized society, these systems face several challenges in adapting to the diverse needs and expectations of individuals and communities. These challenges can be categorized into three aspects: description, prediction, and prescription [85].

- **Description:** Centralized systems often struggle to adapt to the highly heterogeneous and dispersed data found in a decentralized society. These systems may fail to obtain timely and efficient information, which is crucial for making agile decisions and adjustments. In a decentralized context, data sources and information are more diverse, requiring more robust data collection and analysis methods.

- **Prediction:** In centralized predictive models, it is difficult to accurately adapt to the differences between local areas within the transportation system, since each area may have unique characteristics and requirements that a centralized model cannot fully capture. Consequently, predictions made by such models might not be as accurate or useful for all regions, leading to suboptimal transportation planning and resource allocation.
- **Prescription:** Centralized methods lack the precise and effective control, which are needed to address the diverse demands of a decentralized society. The one-size-fits-all approach may result in inadequate solutions for specific situations or regions, and the implementation of these solutions may be delayed due to the bureaucratic nature of centralized systems. This can lead to decreased efficiency and satisfaction among users of the transportation system.

The challenges faced by centralized transportation systems in a decentralized society highlight the need for innovative technologies and approaches that cater to the diverse needs of individuals and communities. ITS in the era of Transportation 5.0 effectively utilizes communication, information technologies, and AI for efficient data-driven traffic management, monitoring, prediction, and behavior analysis. By employing advanced technologies such as AI, Blockchain, DAO₂, and privacy-preserving computing, ITS addresses the description, prediction, and prescription aspects of challenges through the implementation of Privacy Data Collaboration, Transportation Foundation Models, and Decentralized Autonomous Operation and Organization.

Specifically, the widespread use of AI in ITS applications such as autonomous driving, traffic signal control, traffic flow forecasting, demand modeling, driving behavior analysis, and traffic incident monitoring enhances the efficiency and effectiveness of transportation systems in a decentralized context. One example is the use of cooperative driving automation for behavior between vehicles, infrastructure, and other road users to improve overall system performance ([86], [87], [88], [89], [90], [91]). These applications can be deployed in a decentralized system that is powered by Blockchain technology, which is usually referred to as DAO₂. The DAO₂ can manage a wide range of functions of complex and sophisticated systems through Smart contracts, which are executed automatically without centralized control, transparently recorded on a Blockchain, and can be incentivized through the use of characterized tokens. In addition, privacy-preserving computing further strengthens the entire system by allowing for data processing and analysis without compromising the privacy of individuals. It can help to improve the security and privacy of data, guaranteeing that data is kept private while still enabling computation and analysis. By combining these technologies, the development of Transportation 5.0 can evolve towards TRUE₂, which stands for the four important indicators of Trust (T), Reliable (R), Usable (U), and Effective + Efficient (E).

A. Privacy-Preserving Data Collaboration

In the era of Transportation 5.0, the development of hardware, algorithms, and software systems centered on data

technology has opened up new opportunities for ITS research and development [92]. Traffic data collection has moved beyond physical space and into cyber and social space, including travel service platforms, social media, and portable mobile devices. The increase in data volume and complexity has stimulated cross-disciplinary research and the emergence of technologies such as data mining, natural language processing, deep learning, and others.

In the past, traffic data collection was primarily focused on measuring the volume of cars and other vehicles on roads and highways. This was done using physical sensors and cameras that were installed at various points along the roadways. Traditional data collection methods in physical and cyber space have limitations in terms of the space covered and the ability to capture the entire traffic process and environmental variables. With the rise of digital technology and social media, traffic data collection has expanded to include monitoring of social activity as well, such as tracking the number of visitors to location-based services. The data collected from social space can assist in addressing challenges in physical and cyber space data collection [93]. Social space data can provide valuable insights into people's transportation perceptions and behaviors and improve the accuracy of traffic prediction models.

Social data collection, combined with privacy-preserving computing, blockchain, and distributed computing, can improve transportation data utilization in a broader social context. DAO-based city management provides an efficient method for optimizing data utilization and integrating it with data from other subsystems [94]. This could lead to a better understanding of urban dynamics and the development of a multi-system simulation based on data. Implementing a data pricing mechanism could encourage even more data sharing and improve data circulation in a variety of fields. Blockchain technology could result in a secure and decentralized data platform, increasing trust and authenticity in data transactions and improving system security.

In summary, the emergence of advanced technologies such as privacy-preserving computing, blockchain, and distributed computing in the cloud computing and AI era opens up new opportunities for privacy data collaboration. By incorporating these technologies into ITS, we can overcome data heterogeneity, real-time processing, and scalability challenges. This could be helpful in the creation of a data-driven multi-system simulation and the optimization of urban resource allocation, resulting in the greatest possible social benefits.

B. Traffic Foundation Models

Simulation techniques are commonly used in transportation planning and control decisions, with traffic simulation technology widely utilized to study the operational mechanisms of cities. Traffic simulation models are built upon the traffic foundation model, utilizing models of individual agents or groups of agents (such as vehicles or pedestrians) that interact with the infrastructure in the simulated environment. To capture the complexities of urban cities accurately, it is crucial to analyze diverse road user behaviors. The Next Generation Simulation (NGSIM) datasets, provided by the

USDOT, offer valuable real-world data for studying and understanding the interactions among different road users. The work by Xia et al. [95] has contributed to extracting an enhanced version of the NGSIM dataset. However, urban cities are complex and dynamic social systems, where the development of the traffic system is not only the interaction between various traffic players but also changes in social structure and form, presenting new challenges for the future development of simulation technology.

Various simulation models, including macroscopic, microscopic, mesoscopic, and vehicle-level simulation, have been developed to address the challenge of accurately simulating complex large-scale behavior in traffic [86], [96], [97], [98], [99], [100]. These models simplify the system by focusing on a single scale, providing different levels of detail and insights into traffic behavior. In addition, traffic simulation models that depend on rule-based methods utilizing pre-existing models, algorithms, and rules find it challenging to manage high-complexity large-scale agent behavior [101]. As urban systems become more complex, it becomes increasingly difficult to simulate real-time traffic responses dynamically, necessitating more advanced and adaptive simulation methods [102].

In fact, traffic simulation systems can be considered fundamental models for a variety of traffic-related tasks. By using computer algorithms and mathematical models to replicate the behavior of traffic systems, these simulations provide valuable insights into the traffic flow, congestion patterns, and the impact of various transportation policies and infrastructure changes. Data-driven traffic simulation models can serve as pre-trained models for tasks such as traffic flow prediction and traffic signal control. These models have been successfully applied in various fields, including visual and natural language processing, and general decision-making.

By applying cutting-edge methods such as deep neural networks, graph learning, and multi-source data integration, we could turn traffic simulation into Transportation Foundation models, and transform task-specific intelligence into scenarios-oriented generic intelligence. The traffic foundation model is a theoretical framework used to understand traffic phenomena and solve scenario-based problems [94]. Fig. 3 shows one of these models, *TengYun*, which is designed and developed with parallel learning and federated intelligence in TransVerse [103]. It addresses the challenge of insufficient, contradictory, and inconsistently distributed data by implementing data pre-processing, entity alignment, and scenario engineering [104]. It also adopts a “pre-training + prompt tuning” approach to training, which significantly reduces the computational resources and storage required for fine-tuning on edge servers while increasing the utilization of traffic knowledge [105]. *TengYun* provides a solution to the challenge of integrating and utilizing transportation data for efficient and effective management and operation.

TransWorld is an innovative traffic simulation model originally developed by the CAST Lab with the primary objective of constructing Artificial Transportation Systems (ATS) through agent-based technology and object-oriented programming [106]. The model simulates social and behavioral patterns using autonomous agents, which are guided

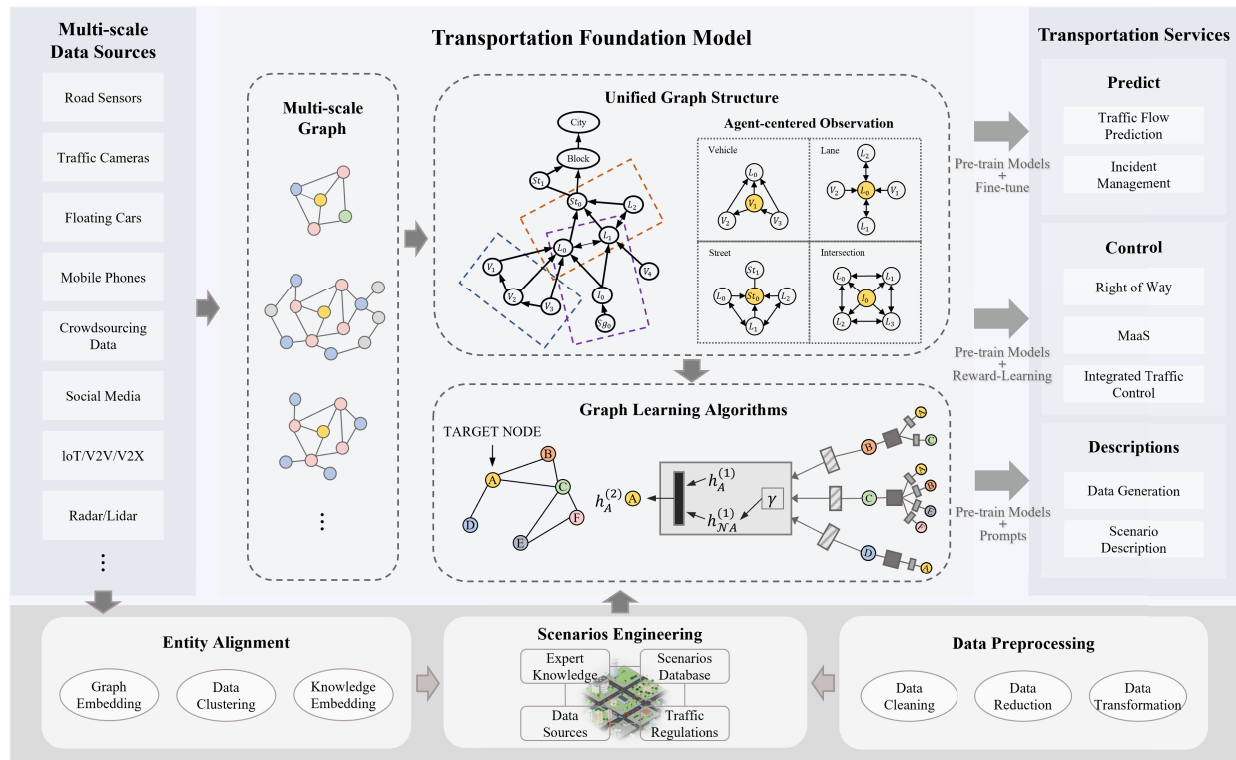


Fig. 3. Conceptual framework of the Transportation Foundation Model for cognition, operation, and management. The model incorporates advanced techniques, such as deep neural networks, graph learning, and multi-source data integration, to facilitate scenario-oriented, generic intelligence in traffic simulation.

by population statistics and behavioral models. Its application proves especially valuable in verifying and validating transportation operations, particularly at the logic level. Through agent interactions, TransWorld can provide a comprehensive description of traffic scenarios at various resolutions. The future of transportation modeling is expected to embrace advanced technologies and methodologies. These upcoming models must possess scalability, flexibility, and expandability to effectively manage the exponential complexity growth of transportation systems. Furthermore, future traffic models should be data-driven, serving as foundational tools to tackle intricate transportation problems by leveraging real-world data [107], [108].

C. Decentralized Autonomous Operations and Organizations (DAO₂)

ITS is a complex system comprised of several subsystems that include multiple participants. These subsystems function interactively to satisfy multitudinous demands such as commuting and tourism. Upon a sophisticated understanding of the system complexity of ITS through the data collection/coordination, modeling, and simulation, the behaviors of participants in ITS can be predicted, whereas the ultimate goal is to manage and control the entire system. The control of ITS aims to achieve a safer, more efficient, and sustainable traffic system by balancing traffic demands and the infrastructure capacity, with the consideration of comfort, safety, emission, and energy consumption requirements [53], [109].

Traffic management and control systems have made significant progress in recent decades to improve the performance

of traffic infrastructures and vehicles through the development of various effective traffic control methods. These control methods can be generally classified based on the response time (e.g., from milliseconds to days) and controlled objects (e.g., regarding infrastructures or vehicles) as depicted in Table I.

Accurate measurement and estimation of traffic states play a crucial role in enabling more efficient, safer, and sustainable traffic control strategies [110]. Existing control methods are theoretically mature and have evolved towards more expeditious, agile, and adaptive with the aid of AI in the era of Transportation 5.0. However, the complex nature of ITS has impeded these methods from functioning as intended in practice and enforced the entire system into the Braess paradox while implementing these methods without coordination [111], [112]. For instance, the green-wave traffic signal control may induce traffic congestion on uncontrolled roads, reduce the network capacity, and ultimately deteriorate the overall traffic efficiency [113]. Therefore, investigating integrated and coordinated control methods with the consideration of participants in multiple ITS subsystems and scales, as well as with multiple objectives renders a non-trivial challenge to be tackled.

Recent studies have made some efforts on multi-scale and multi-objective integrated traffic control, which have suggested its efficiency in mitigating traffic congestion, guaranteeing safety, and enhancing sustainability. Typical studies include the integration and coordination of multiple infrastructure control methods within one specific subsystem (e.g., on highways) [114], [115] or in multiple subsystems (e.g., on highways and urban roads) [106], multiple connected and automated

TABLE I
TRAFFIC CONTROL MEASURES IN THE ERA OF TRANSPORTATION 5.0

Planning Horizon

Days

Hours

Minutes

Seconds

Milliseconds

Roadside Traffic Control	City Level	Network Level		Link Level	Lane Level	-	-
	Traffic toll, etc.	Route guidance, Network congestion warning, etc.	Perimeter gating control, etc.	Traffic signal control, Ramp-metering, Variable speed limit, etc.	Lane prohibition, Reservable flow lane, etc.	-	-
Vehicle Control	-	Strategic Level	Maneuver Level				Action Level
	-	Vehicle routing, etc.	ADAS, CAV platoon organization, etc.			CAV Lane allocation, CAV desired headway control, etc.	Motion planning

vehicles [90], [116], as well as between road infrastructures and vehicles [117], [118]. These studies are conducted either in a centralized or a decentralized scheme. The centralized scheme normally assumes perfect knowledge of all the controlled participants' states, based on which a generalized objective is formulated summarizing the objectives of each controlled participant. On the contrary, under the decentralized scheme, the involved participants are controlled based on their own objectives, which are correlated to the observation and prediction of their connected participants' states. While the centralized scheme is struggling with its scalability, stability, and security, which makes it incompetent of handling unlimited participants with heterogeneous attributes (e.g., different scales, response times, control objectives, etc.); control methods adopting the decentralized scheme also leave much to be desired with their compromise on the optimality [119]. Fortunately, the emergence of Blockchain technology and DAO₂ has shed some light on solving the issue of developing integrated and coordinated traffic control systems in a trustable, reliable, useful, and efficient manner [120], [121], [122], [123].

By implementing a system based on DAO₂, coordinated control across multiple participants and subsystems can be achieved through a series of smart contracts [94]. Specifically, the absence of centralized authority supported by Blockchain technology offers security and transparency, as well as simplifies and secures the sharing among a range of traffic data from different subsystems, including vehicle location, travel speed, traffic signal, etc.; Smart contracts can further process these data automatically and enabling appropriate actions to enhance traffic conditions. In general, this approach enables community-driven autonomy that aligns with the principles of mutual benefits, transparency, and safety [124].

For example, traffic signals can be dynamically controlled based on DAO₂ by a series of smart contracts that optimize traffic flow and reduce congestion based on real-time data, such as vehicle location and speed [125]. These smart contracts can also facilitate the objectives of other participants in different subsystems, such as the safety of pedestrians and cyclists, by monitoring their behavior using real-time sensor

data and taking appropriate actions, such as modifying traffic signals, to achieve mutual benefits. Furthermore, Blockchain technology guarantees secure and transparency within ITS. For instance, while implementing a smart ticketing system for public transportation, the DAO₂-based integrated control approach can effectively reduce ticket fraud and decrease administrative expenses, meanwhile facilitating other objectives, such as reducing traffic congestion, through decentralized and tamper-proof record management.

In summary, the combination of ITS and blockchain technology can achieve more efficient, safer, and sustainable traffic control. Through integrated and coordinated control via Blockchain technology and DAO₂, the traffic system can indeed better meet people's travel demands while reducing traffic congestion, improving safety, and contributing to the sustainable development of cities.

D. Welcome to HANOI

The three cornerstone technologies, namely Privacy-preserving Data Collaboration, Traffic Foundation Models, and DAO₂, provide support for the future ITS from the perspectives of description, prediction, and prescription, shown in Fig. 4. These technologies are also prepared to meet the technical and social architecture requirements of Transportation 5.0, the decentralized society. Through Privacy-preserving data collaboration, information can be collected from dispersed sources, and the Foundation Model can form a unified consensus from fragmented social organizations. Finally, the DAO₂ align the interests of all parties to maximize overall social benefits. Compared with Transportation 4.0, Transportation 5.0 involves a significantly increased number of participants. Therefore, the technical requirements are broader, including IoT, blockchain, and intelligent technologies.

In essence, the core of these three technologies is to achieve the coalescence of social consensus. From the most basic consensus of information disclosure methods, gradually developing into the consensus of overall status cognition, forming a consensus on the behavior of each subject in local time and space, and ultimately precipitating into a series of

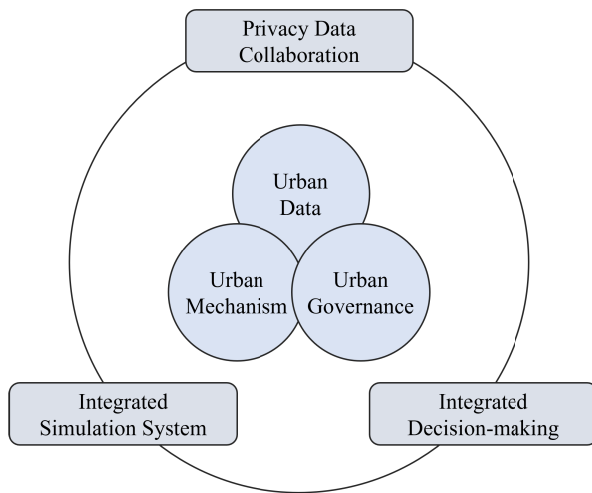


Fig. 4. Illustration of the three cornerstone technologies, namely Privacy Data Collaboration, Integrated Simulation System, and Integrated Decision-making, which play a crucial role in shaping the future of ITS.

long-term consensus on behavior norms and resource exchange ratios. The first three consensuses often play a role in the social subsystem of transportation, while the last long-term consensus will have an impact beyond transportation, achieving an organic fusion of transportation, and the other social subsystem.

Although there is still much research required on the specific form and implementation path of these consensuses, it is believed that their ultimate goal is to achieve the TRUE₂ of the overall system, that is the four important indicators of Trust (T), Reliable (R), Usable (U), and Effective + Efficient (E). It also aims to ensure safety in the physical world (Safety Index), security in the information world (Security Index), sustainability in the ecological world (Sustainability Index), optimization of individual needs and rights (Sensitivity Index), high-quality services for everyone (Service Index), and intelligent infrastructure for everyone (Smartness Index) in the 6S society. Therefore, the “6I” intelligence is required, including Cognitive Intelligence and Parallel Intelligence at the technical level, Crypto Intelligence and Federated Intelligence at the operational and management level, and Social Intelligence and Ecological Intelligence at the organizational level [126]. This intelligence will ultimately form HANOI, which is a combination of human intelligence (H), artificial intelligence (A), natural intelligence (N), and organizational intelligence (OI) [127]. Fig. 5 explains the six types of intelligence and indexes that form HANOI.

V. FUTURE SCENARIOS AND REQUIRED SYSTEMS

In addition to cornerstone technologies and tool systems, new traffic modes such as Mobility as a Service (MaaS) and autonomous driving are emerging to accommodate alterations in social structures. The future ITS scenarios will guide future transportation development.

A. Social-Participated AI Test and Scenario Engineering

The successful deployment and commercialization of Intelligent agents, such as AVs, necessitate extensive testing.

However, the growth of AI tests has been hindered due to insufficient relevant data and repetitive test scenarios.

Inspired by the concept of DAO₂, AI tests should not be constrained to a single type of vehicle or restricted area. Socially participated AI tests should be adopted, in which tests are conducted in diverse contexts with many connected vehicles, conventional vehicles equipped with data collectors, and all traffic participants (including bicycles and pedestrians). SE (Scenario Engineering) is an integrated reflection of the scenarios and activities within a certain temporal and spatial range, where all actionable AI are encouraged to complete the design, certification, and verification [126]. SE shapes the AI systems to be more coherent to their original testing purpose by preventing repetitive scenarios and generating more valuable scenarios. Parallel learning is proposed as a good tool for the AI test and scenario engineering [128]. DeCAST in the TransVerse platform was designed and constructed according to the principle of DAO₂, as shown in Fig. 6 [94]. Through parallel learning, the evaluation and development of AVs could be greatly accelerated. As a case study, Li et al. [129] present how to design reliable intelligence tests for AVs.

It is essential to acquire a significant amount of data for the AI testing method in generating comprehensive scenarios, especially for mixed-traffic environments. Comprehensive scenarios, in turn, can alleviate the bottleneck in developing autonomous driving technology. To accomplish this goal, an integrated testing platform is required to connect the virtual simulation test field, human-driven driving simulator, and actual test field. As a result, it will establish a closed-loop process to develop, test, and advance new technologies. The proposed simulation-driven AI could meet the requirement for the Try-Test-Adjust design policy to build an intelligent machine when integrating virtual with real data collected [130].

B. Trans-Scale Control and RoW-Based Traffic System

The current control range of traffic lights has raised research concerns, as signal lights are placed only at intersections and expressway ramps, leading to potential conflicts. Right of Way (RoW) in traffic interactions could be a crucial new concept and technology in overcoming the challenges associated with traditional traffic control.

To fully utilize limited road resources, an efficient traffic control system must address the following challenges:

- Precisely determining the arrival time of vehicles at intersections to deploy appropriate control signals;
- Ensuring that the distribution and communication of RoW are seamlessly integrated throughout the entire driving process;
- Accurately measuring and controlling the movement of each vehicle on the road.

The emergence and advancement of V2X technology provide new opportunities to address RoW ownership issues. It creates a network connecting humans, vehicles, roads, and other transportation elements, allowing for ubiquitous information sharing. However, the transportation system faces

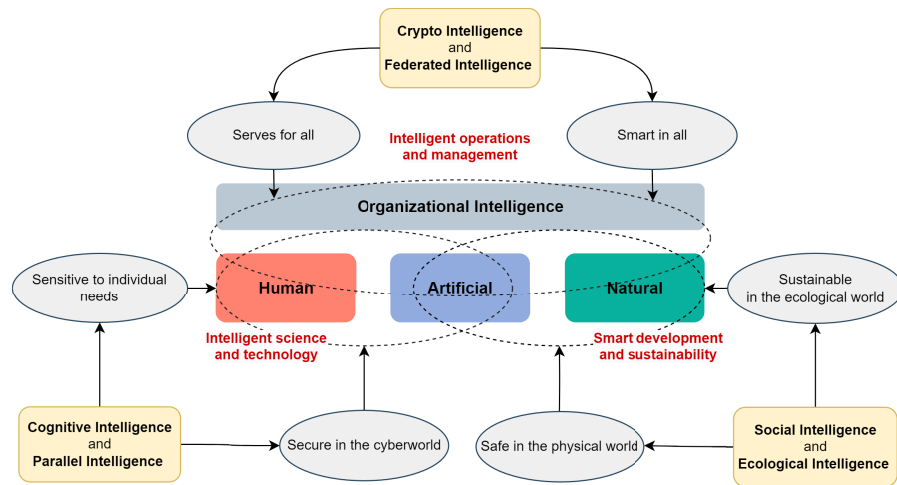


Fig. 5. The six types of intelligence and indexes that form HANOI. This figure shows the different types of intelligence and corresponding indexes that contribute to the ultimate goal of achieving Trust, Reliable, Usable, and Effective + Efficient (TRUE₂) transportation systems. HANOI represents the combination of human intelligence (H), artificial intelligence (A), natural intelligence (N), and organizational intelligence (OI) to ensure the overall success of the transportation system.

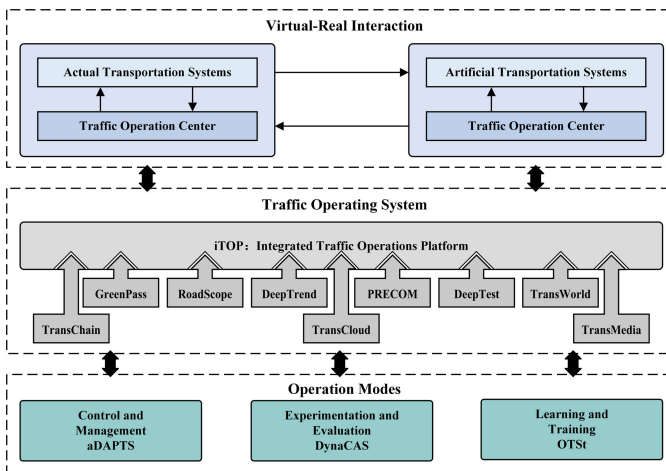


Fig. 6. The platform architecture of DeCAST: Parallel transportation management and control.

challenges due to the absence of a unified control objective for coordinated subsystem control. A new concept called “Generalized Right of Way” is required to address this issue in traffic systems across various agents at different scales. As a general usufruct of the transportation system, the RoW is a fundamental attribute granting permission to occupy specific travel resources for a specific period. By utilizing this concept, a trans-scale traffic control system can be constructed to address the issue of fragmented control strategies.

With the continuous development of urban intermodal transportation demand, the limitations of single scale control methods gradually emerge, restricting the ability to integrate multi-scale schemes to achieve control objectives. Recently, some studies have been conducted to explore the cooperation of trans-scale traffic control tasks. As an important issue in ITS, how to develop a unified control mechanism from the standpoint of maximizing social resource allocation and appropriately coordinating the employment of multiple

control modes still requires further in-depth research. The RoW facilitates seamless transformation between different subsystems and scales within the traffic system by quantifying one aspect’s impact on all others. Based on the unified control objective (RoW), a multi-agents collaboration mechanism for trans-scale traffic control can be achieved. The mechanism allows for a multi-agent joint control scenario, due to its development in accordance with the percentage of resource consumption and control effectiveness of each agent.

The RoW can also be exchanged with currency, carbon emissions, energy consumption, and other rights/interests to achieve a tradable system. By connecting the transportation subsystem with the carbon sink trading system and LBS system, this system can improve resource allocation based on data integration from users and various travel resources. Then, by collaboratively controlling all elements in a complex traffic system and abstracting numerous social interactions as generalized usage rights, a unified quantitative method for resource recycling mechanisms can be provided. Furthermore, the cross-system decision-making scheme to optimize the overall social benefits can be used to allocate social resources.

As one of the best practices of traffic control based on RoW theory, an application of parallel transportation system, Traffic Operation Recommendation for Controlling Hangzhou Road Networks (TORCH), has been successfully deployed on the Hangzhou road network, realizing city-level traffic control at more than 1,000 intersections while enhancing the capacity of handling traffic congestion incidents by two to three times, as shown in Fig. 7.

We have also investigated proactively preventing collisions using a set of decision criteria for AVs to effectively determine the RoW [131]. In order to boost driving safety for AVs, we have identified three key concepts and developed a rule-based, nine-step communication-decision paradigm, as shown in Fig. 8. The decision criteria are intended to be clear and quickly computed with the least amount of communication so that human drivers and AVs can comprehend each other’s behaviors and intents.

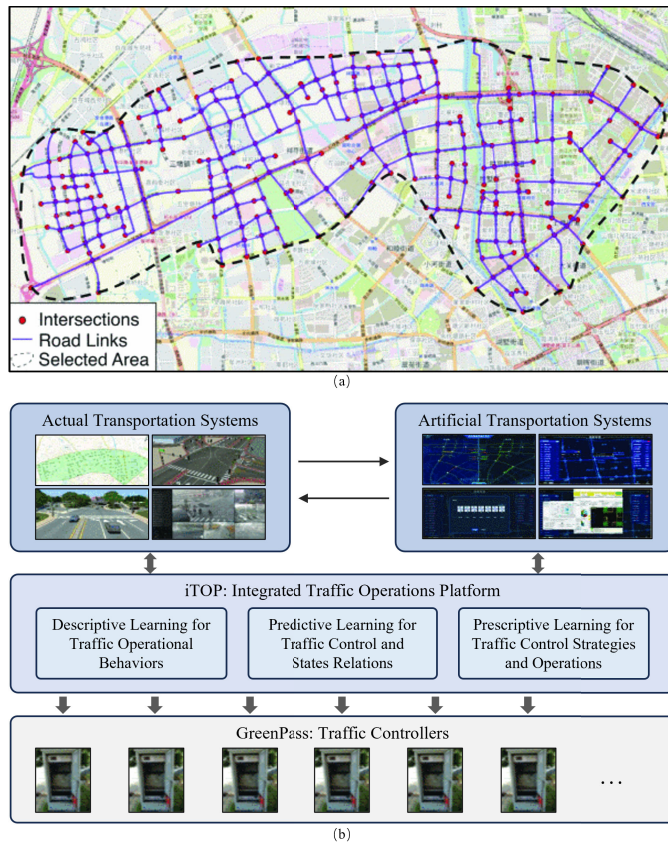


Fig. 7. Traffic Operation Recommendation for Controlling Hangzhou Road Networks (TORCH) is an exemplary application of a parallel transportation system that demonstrates one of the best practices in traffic control based on the concept of RoW theory. (a) The case study area of TORCH, and (b) The architecture of TORCH.

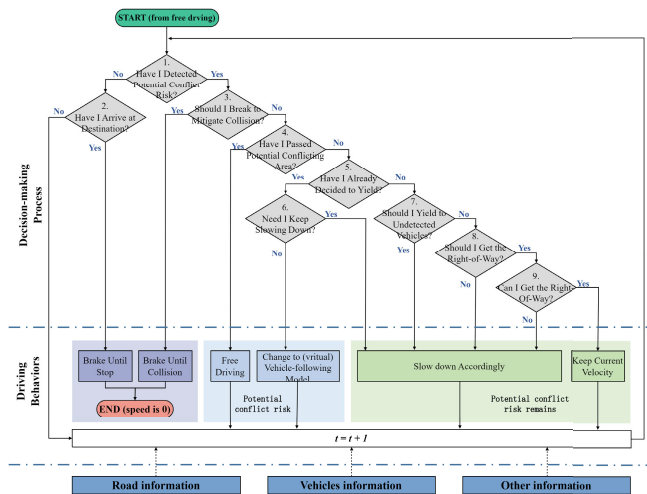


Fig. 8. Flowchart of the communication-decision model for RoW in proactive collision prevention for AVs. The model outlines a rule-based, nine-step paradigm that facilitates clear and efficient communication between AVs and human drivers, ensuring safe and coordinated interactions on the road.

C. MaaS and Human-Oriented Operating Systems

MaaS is an emerging transportation business model that assembles the services of different mobility service providers through exchanging data, simplifying intermediation, and providing customers with seamless service and a unified payment

interface through a continuous pipeline. While there are multiple definitions of MaaS, most share a common vision of transforming the need for ownership of vehicles to enable users to access the most desirable shared mode of transportation on a trip-by-trip basis. Several cities worldwide have already adopted the concept of MaaS, numerous practical studies on driving MaaS implementation [132] and research examining the business models of MaaS [133] have been published in recent years. For city administrators, MaaS provides an opportunity to build and operate city-scale transportation systems and services in a holistic manner, rather than managing each transportation service independently.

A completely new system is needed to integrate all aspects of transportation into social life. Such a system will present a unified, human-oriented interface, coordinating with people's behavior habits and preferences, and use human feedback to improve its underlying agent system. Some researchers suggest that such a system could be called a Human-oriented Operating System (H2OS) [134]. Like the operating system for computer software and hardware, H2OS is a complex system composed of multiple components that can virtualize transportation resources, parallelize transportation requests, and apply transportation planning to real-world operation.

The H2OS for transportation environment includes multiple components, such as traffic data collection, processing, and storage modules, traffic request management modules, resource scheduling modules, etc. Through component collaborations, H2OS can virtualize and optimize transportation resources, making transportation smarter and more efficient. A core feature of H2OS is its unified interface through which users can easily obtain transportation resources and perform planning and scheduling operations.

In the design of H2OS, a decentralized architecture should be used, which provides numerous benefits. For instance, it makes H2OS more resilient to local failures or attacks. Additionally, as new nodes join the network, they can contribute their resources to the system, enabling H2OS to scale more easily. This architecture can also enable H2OS to allocate and optimize transportation resources more efficiently, as it can take advantage of distributed computing and real-time data processing to make better decisions. Furthermore, privacy and security can be improved as the users' data is distributed across the network rather than being stored in a central location. This can also make the system more transparent and accountable, as all network participants can view the system's transactions and operations.

VI. FUTURE AND VISION

In conclusion, it is important to emphasize the ongoing paradigm shift in ITS from a technology-centric focus to a more society-centric one. This transition is being driven by adopting Cyber-Physical-Social Systems architecture and parallel systems methodology, which offer innovative and comprehensive solutions to pressing transportation challenges. These approaches are being rapidly developed and implemented across three key applications: Privacy-preserving Data Collaboration, Traffic Foundation Models, and Decentralized Autonomous Operation and Organization.

The success of past and present ITS practices validates the position and related assertions of ITSS TC for Transportation 5.0 proposed a decade ago. As the development of ITS continues to evolve, scenarios such as social-participated AI Test, ROW-based traffic control, and MaaS will play a crucial role in advancing ITS towards a more society-centered architecture. This significant shift will not only transform the ITS landscape but also profoundly impact the organization of related fields, such as urban planning, energy, and environment, steering transportation systems toward human welfare and sustainability-centered objectives.

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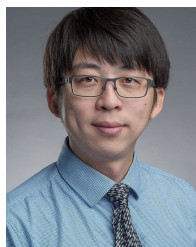


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